
CISUC

Fair Causal KAN

Bayesian splines, surfaces and screening

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Current implementation · November 2026

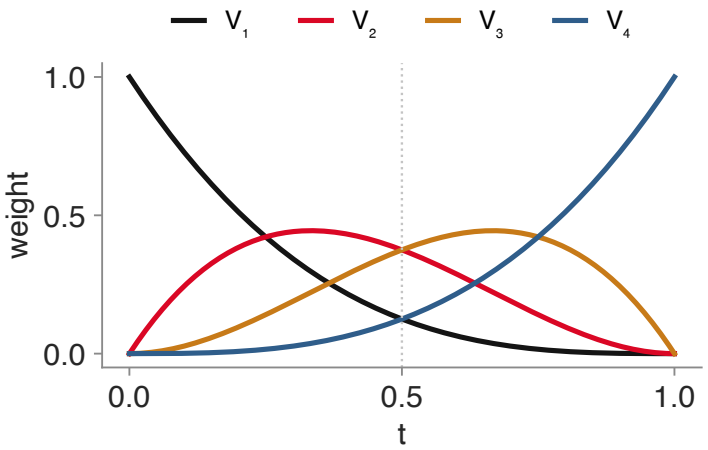
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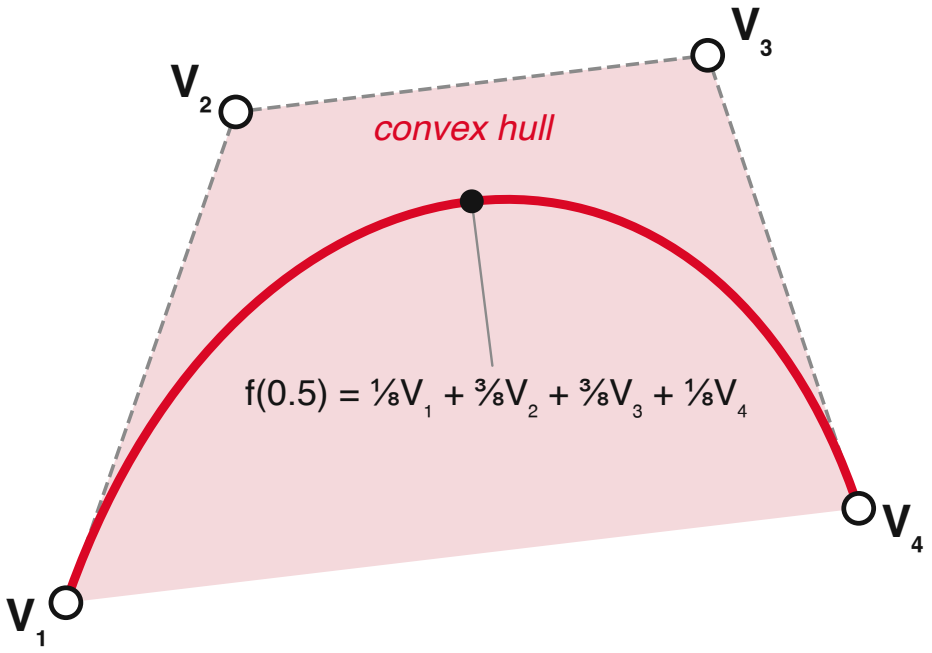
Bézier curves: a curve drawn from four points

As t goes from 0 to 1, the point $f(t)$ travels from V_1 to V_4, pulled towards V_2 and V_3 on the way.

$$f(t) = (1-t)^3\, V_1 + 3t(1-t)^2\, V_2 + 3t^2(1-t)\, V_3 + t^3\, V_4$$

The four weights, as t goes from 0 to 1:





No unknown constants

Weights depend only on t.

Inside the hull

Weights ≥ 0, adding up to 1.

Ends fixed

Starts at V_1, ends at V_4.

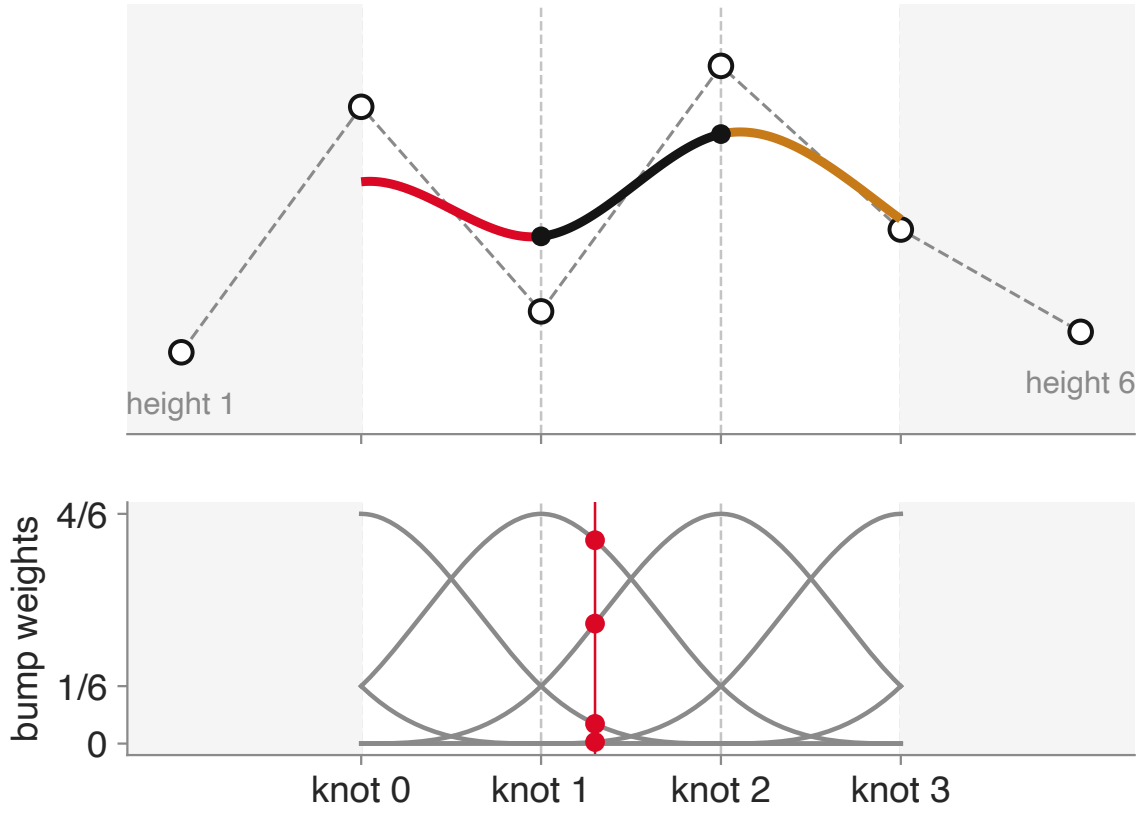
B-splines: Bézier pieces joined without a kink

B = basis. Every curve is built from the same set of bumps (the basis), each scaled by its own height:

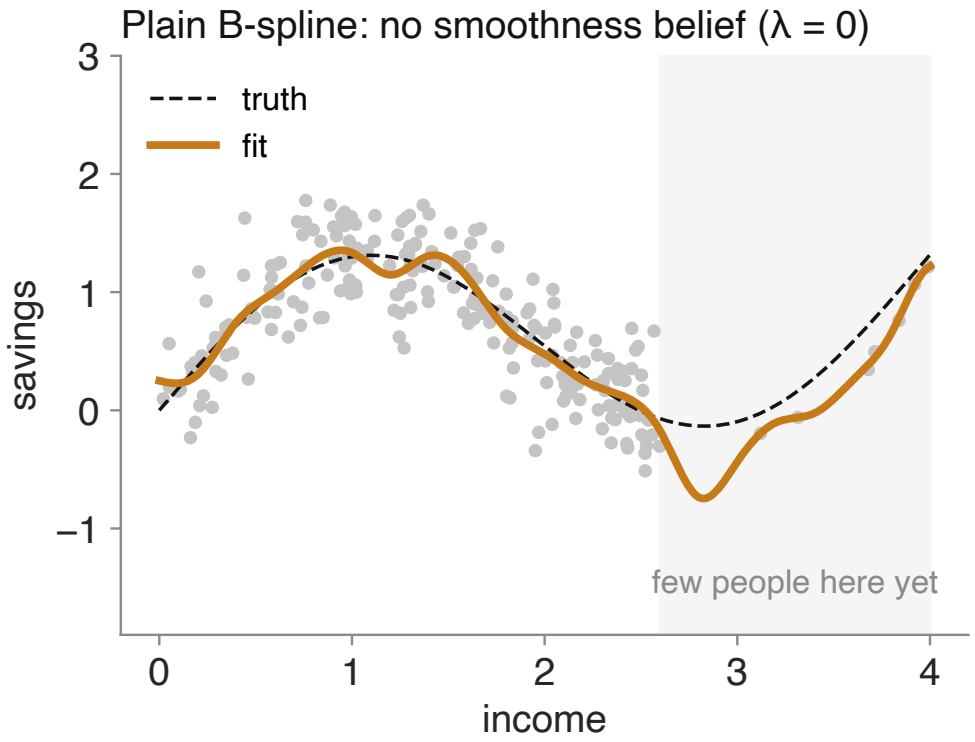
$$f(x) = \sum_k B_k(x)\, h_k$$

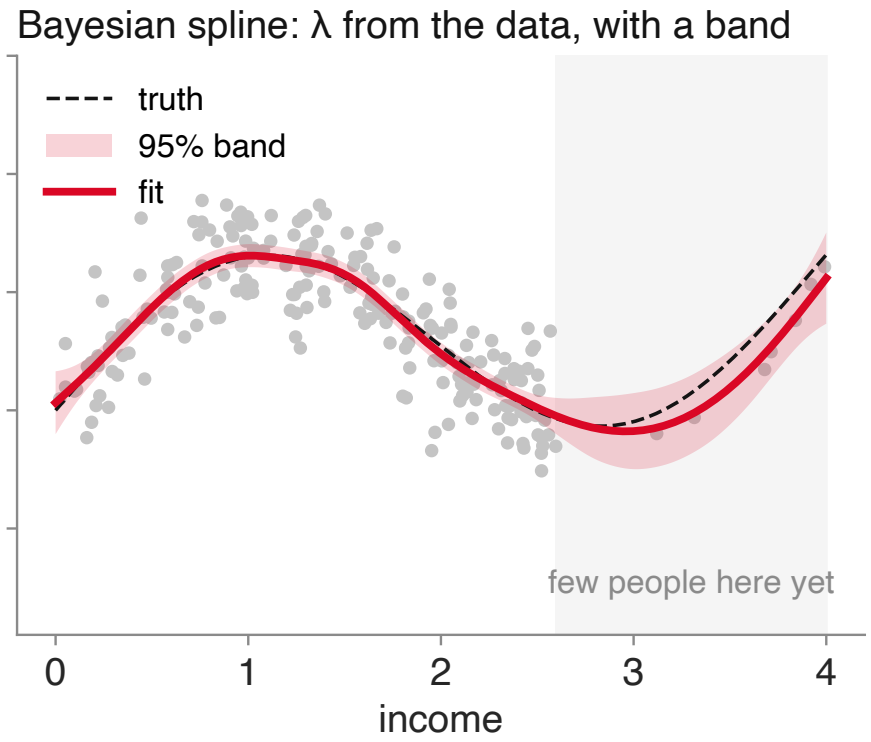
Between two knots, four bumps are active, so each piece is a cubic Bézier curve. Neighbouring pieces share three of their four heights.

1. **Smooth joins**
   At a knot, value, slope and bending all match.
2. **Local**
   One height moves the curve over at most 4 intervals.
3. **Inside the hull**
   Weights never negative, adding up to 1 (at a knot: $\frac{1}{6}, \frac{4}{6}, \frac{1}{6}$).
4. ***n* intervals (gaps between knots), $n + 3$ heights**
   In kairos: `Spline(low, high, intervals)`.



Plain B-splines in a drifting world





No band
Change or noise?

Wiggles
where few people are

No smoothness belief
$\lambda = 0$

Bayesian splines
band + λ from the data

Bayesian splines (1/2): believability before and after the data

1 **Believability before the data**

believability$(h) \propto \exp\left(-\frac{\lambda}{2\sigma^2}\sum_k (h_{k-1} - 2h_k + h_{k+1})^2\right)$

Heights 1, 2, 3 do not bend (0): most believable. Heights 1, 3, 1 bend by -4: cost $16 \cdot \frac{\lambda}{2\sigma^2}$. λ is how strongly we believe the curve is smooth.

2 **After the observations**

believability after $\propto$ fit to the data $\times$ believability before.

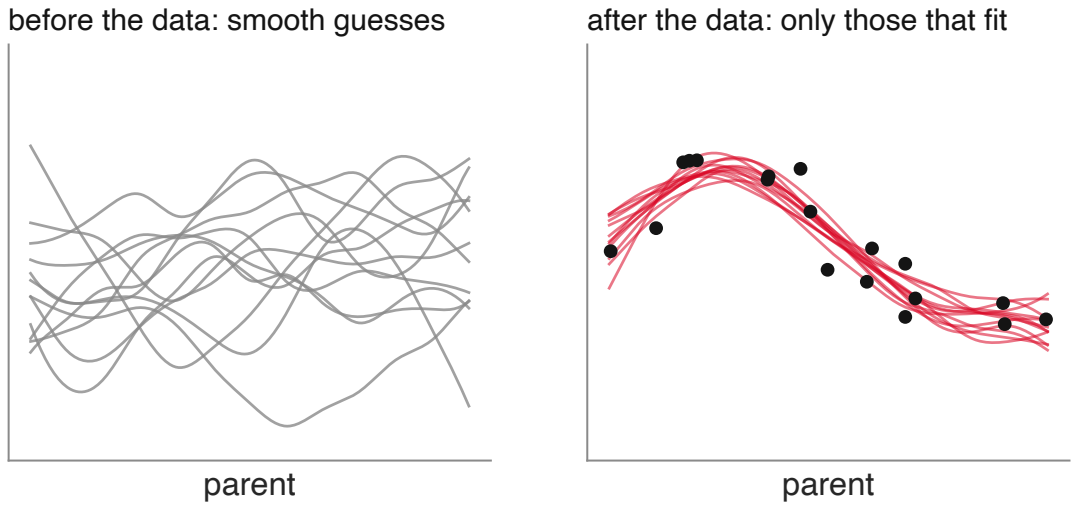
Most believable heights: $(W^\top W + \lambda S)\, h = W^\top y$

W: each person's bump weights (one row each) · S: the bending cost

3 **How unsure the curve is**

$\Sigma = \sigma^2 (W^\top W + \lambda S)^{-1}$ gives each height's spread.

Band = curve $\pm$ 1.96 spreads (about 95% of believable curves).



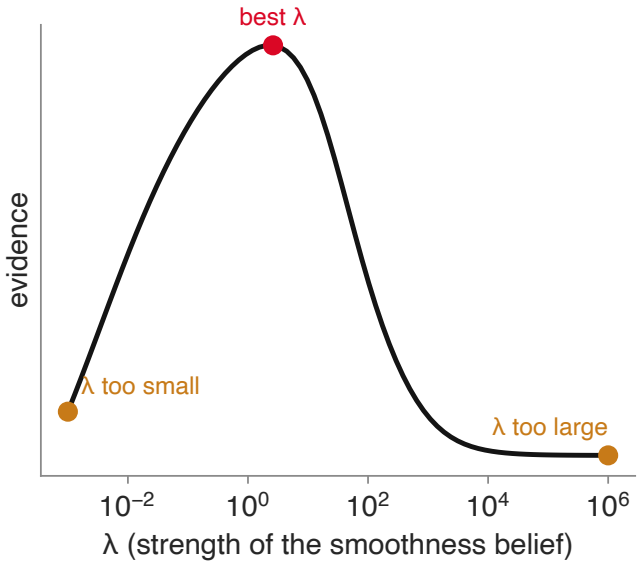
Left: random curves that are believable before any data (no bending is cheap). Right: random curves still believable after 18 people; together they form the band.

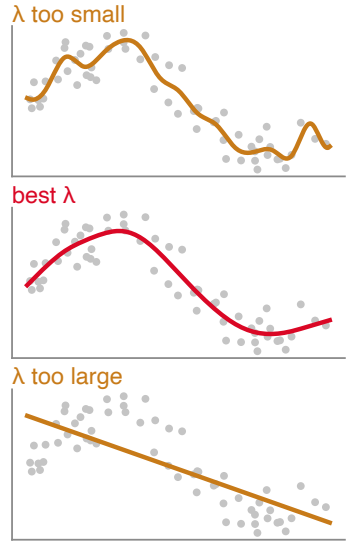
Bayesian splines (2/2): choosing λ and the noise size

1 **Evidence: how probable is the data under λ?**

Averaged over every believable curve. Too small λ: many wiggly curves share the believability, so each fits little. Too large: only straight lines, which miss the data.

$\log \text{evidence} = -\frac{n}{2}\log\sigma^2 + \frac{1}{2}\log|\lambda S| - \frac{1}{2}\log|W^\top W + \lambda S|$





2 **Noise size σ^2**

The average square of what the curve leaves unexplained:

$\sigma^2 = \left(\sum_i (y_i - \hat{y}_i)^2 + \lambda h^\top S h\right)/n$

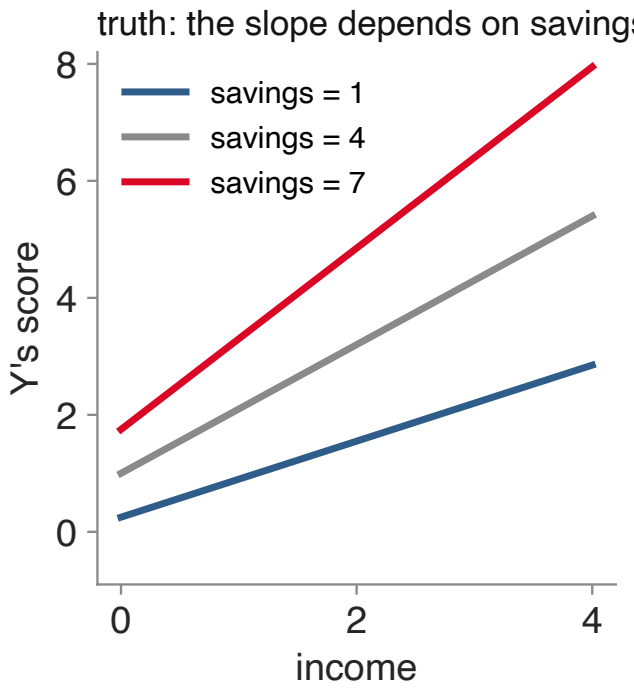
Leftovers $0.3, -0.2, 0.1$:

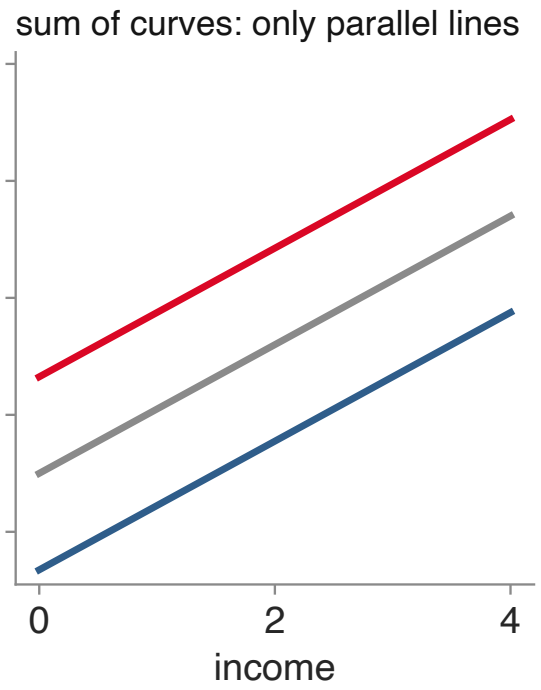
$\sigma^2 \approx (0.09 + 0.04 + 0.01)/3 = 0.047.$

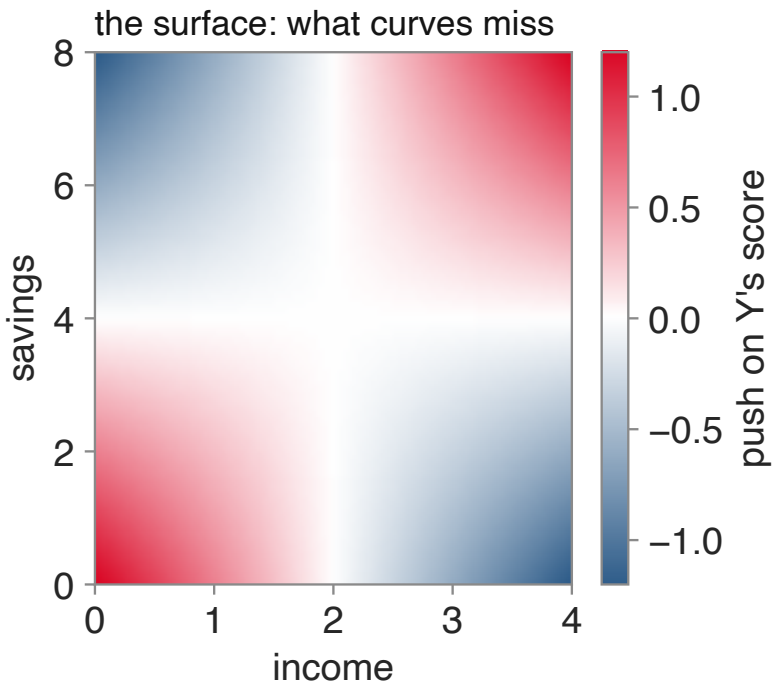
3 **One λ per curve**

Grid from 0.1 to 10^4; each curve's λ is tuned in turn, three rounds. For Y (a chance, not a bell): Newton's method and a Laplace approximation.

Surfaces: when one parent's effect depends on another







The problem

Curves only shift lines.

The fix

A grid of 9 × 9 heights.

Only the interaction

The curves keep their meaning.

Screening: which pairs of parents get a surface?

1. **Fit curves only, keep the leftovers**

   leftover = actual value − what the curves predict. An interaction leaves a pattern that needs both parents.

2. **FAST: score every pair by quadrants**

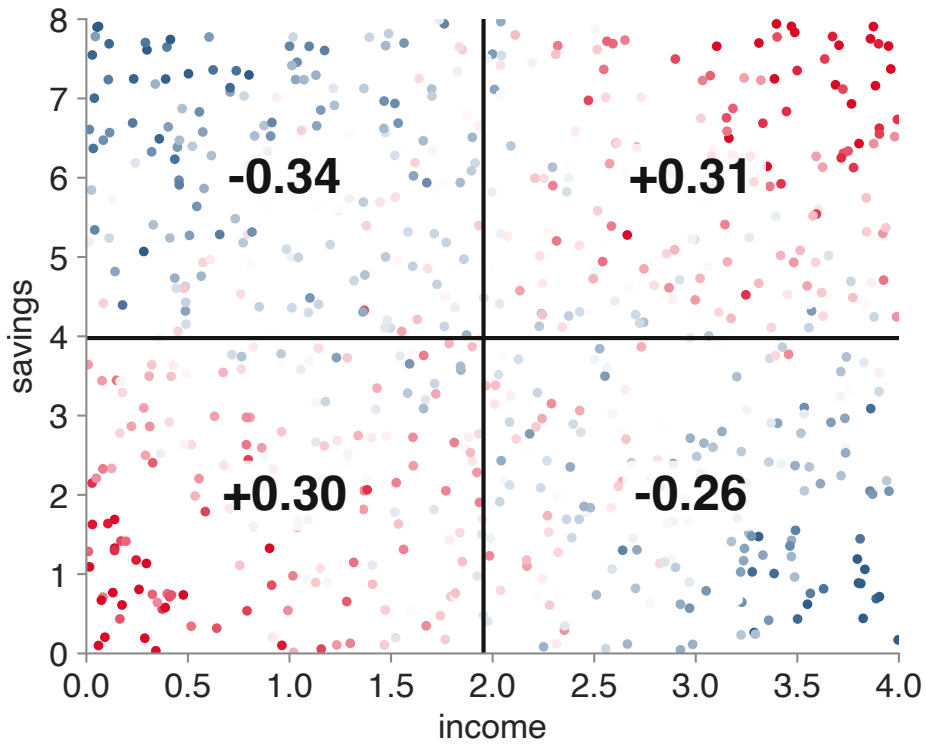
   Cut both parents in two; average the leftovers in each quadrant. Opposite corners with the same sign mean an interaction.

   score $= \sum_q n_q \bar{r}_q^2 \,/\, \sum_i r_i^2$, the share the 4 averages explain

   Cuts at 25, 50 and 75% of each parent: the best of 9 tries is kept.

3. **Check the best three with the evidence**

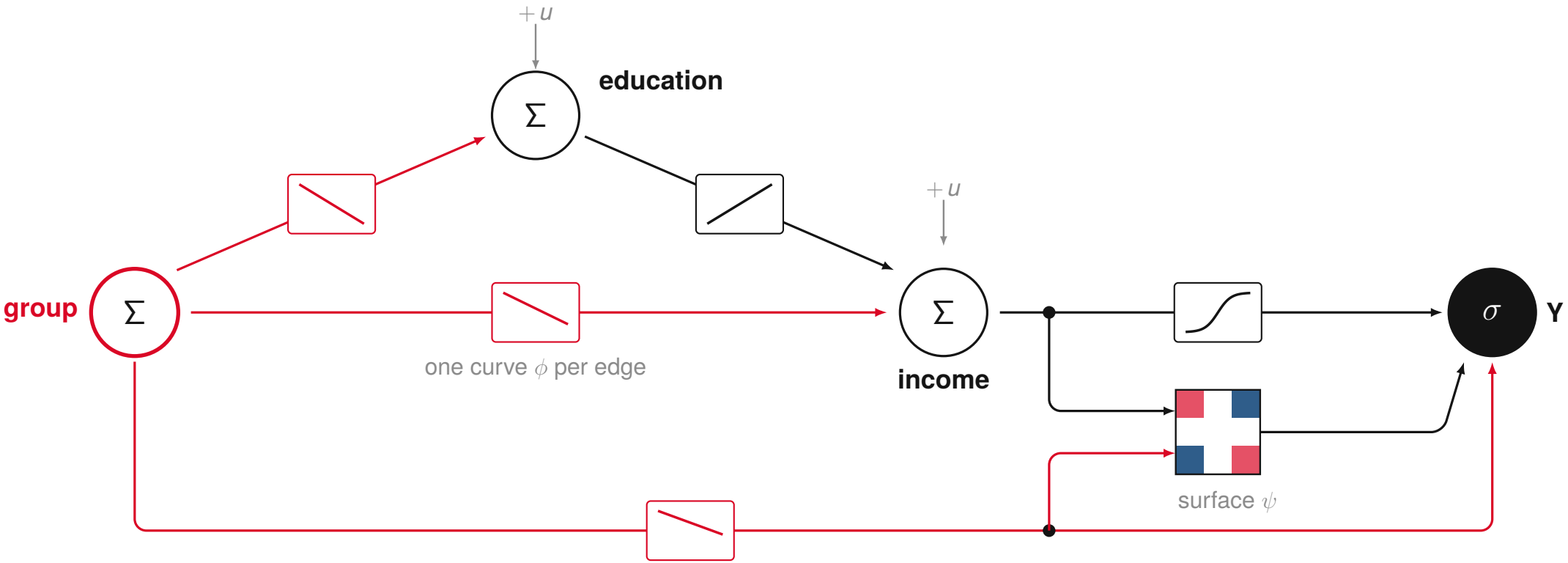
   Refit with the surface added; keep it only if the data becomes more than $e^3 \approx 20$ times more probable.



Leftovers for Y with an income × savings term; the numbers are the quadrant averages (Lou et al., 2013).

The current causal KAN

$$x_j = c_j + \sum_i \phi_{ij}(x_i) + \sum_{(a,b)} \psi_{abj}(x_a, x_b) + u_j \qquad P(Y{=}1) = \sigma\left(c_Y + \sum_i \phi_{iY}(x_i) + \sum_{(a,b)} \psi_{abY}(x_a, x_b)\right)$$



Edge

one B-spline curve, fitted Bayesian

Square

a surface, kept by screening

Red

edges out of the protected group